

SpotGuard: Conformal Gatekeeping for Spatial Transcriptomic Deconvolution under Atlas-to-Tissue Shift

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Abstract

Spatial transcriptomics (ST) deconvolution tools (e.g., `cell2location`, `RCTD`, `SPOTlight`) estimate cell-type proportions across tissue spots, but unadjusted posterior credible intervals suffer severe miscalibration under atlas-to-query distribution shifts. We demonstrate **SpotGuard**, an operational, interactive conformal gatekeeping system for ST deconvolution. SpotGuard formulates nonconformity via Total Variation (TV) geometry on the probability simplex and adapts to distribution shifts via density-ratio weighted conformal prediction. Integrated into a responsive web interface with live miscoverage (α) slider controls, SpotGuard dynamically prunes uncalibrated tissue spots, achieving 89.8% empirical coverage on controlled shift benchmarks where Bayesian joint regions collapse.

Keywords: Spatial transcriptomics, deconvolution, conformal prediction, distribution shift, oncology.

1. Introduction

Clinical Problem & Importance. Spatial transcriptomics (ST) enables spatial gene expression profiling across intact histological tumor biopsies (Marx, 2021; Rao et al., 2021). Because sequencing-based ST captures multi-cellular mixtures (55 μm spots, 5–30 cells), computational deconvolution methods—such as `cell2location` (Kleshchevnikov et al., 2022), `RCTD` (Cable et al., 2022), and `SPOTlight` (Elosua-Bayes et al., 2021)—infer cell-type proportion vectors $\hat{y} \in \Delta^{K-1}$. In clinical oncology, these deconvolution maps guide critical decisions, including quantifying cytotoxic CD8⁺ T-cell infiltration, detecting tertiary lymphoid structures (TLS), and delineating surgical resection margins.

Failure Mode of Current Approaches. Deconvolution models rely on reference scRNA-seq atlases.

When reference atlases and patient tissues diverge due to sequencing platform (e.g., 10x Chromium vs. Slide-seq), tissue preservation (FFPE vs. fresh frozen), or patient demographics, exchangeability fails. Standard Bayesian posteriors become severely overconfident: a clinician cannot distinguish a spot where the model is confident and correct from one where it is confident and wrong. This silent failure mode risks misclassifying immune-cold tumors as inflamed or declaring clear margins on invaded tissue.

Intended Use & Target Population. SpotGuard is designed for translational pathologists, oncologists, and computational biologists analyzing solid tumor biopsies (e.g., breast carcinoma, melanoma, colorectal cancer). SpotGuard operates as an interactive conformal gatekeeper deployed alongside deconvolution pipelines to prevent uncalibrated predictions from entering clinical workflows.

2. Method

Compositional Nonconformity on the Simplex.

Cell proportions inhabit the closed simplex $\Delta^{K-1} = \{v \in \mathbb{R}_+^K : \sum_{k=1}^K v_k = 1\}$. Because sparse proportions cause log-ratio singularities at zero, SpotGuard formulates nonconformity via Total Variation (TV) distance: $\mathcal{S}_{\text{TV}}(x, y) = \frac{1}{2} \|y - \hat{y}\|_1 = \frac{1}{2} \sum_{k=1}^K |y_k - \hat{y}_k| \in [0, 1]$, defined everywhere on the closed simplex.

Pseudo-Spot Calibration Procedure. Because single-cell ground truth is unobservable in physical spatial spots, SpotGuard synthesizes $M = 1,000$ calibration pseudo-spots $\mathcal{D}_{\text{cal}} = \{(x_i^{\text{syn}}, y_i^{\text{syn}})\}_{i=1}^M$ from reference single cells by drawing mixtures $y_i^{\text{syn}} \sim \text{Dirichlet}(\beta)$, aggregating counts $N_i \sim \text{Poisson}(\lambda)$, injecting ambient RNA $\rho \sim \text{Beta}(a, b)$, and matching empirical spatial library sizes.

Weighted Conformal Prediction under Covariate Shift. To correct for atlas-to-query distribution shifts ($P_{\text{ref}}(X) \neq P_{\text{query}}(X)$), SpotGuard trains

Table 1: **Empirical Coverage & Efficiency at 90% Nominal** ($\alpha = 0.10$). Simplex Vol. ($\mathcal{V}$) denotes the mean TV ball radius $\hat{q}_\alpha$ on Δ^{K-1} . Bayesian Joint CI is an unadjusted product of marginal intervals (K -box).

Regime	Method	Coverage [95% CI]	Simplex Vol. ($\mathcal{V}$)
In-Distribution	Bayesian Joint CI	1.0% [0.2%, 2.0%]	0.012
	Bayesian Marginal CI	85.4% [82.1%, 88.6%]	0.041
	SpotGuard (Ours)	92.4% [90.2%, 94.6%]	0.038
Cross-Donor	Bayesian Joint CI	2.0% [0.8%, 3.4%]	0.015
	Unweighted Conformal	89.8% [87.4%, 92.4%]	0.054
	SpotGuard (Ours)	89.8% [87.4%, 92.4%]	0.058
Cross-Platform	Bayesian Joint CI	1.4% [0.4%, 2.6%]	0.018
	Unweighted Conformal	84.6% [81.4%, 87.8%]	0.062
	SpotGuard (Ours)	89.4% [86.8%, 92.1%]	0.071

a regularized logistic discriminator $d_\phi(x)$ on spot gene embeddings to compute density-ratio weights $w(x) = \frac{d_\phi(x)}{1-d_\phi(x)} \frac{M_{\text{ref}}}{M_{\text{query}}}$ (Tibshirani et al., 2019; Barber et al., 2023). We evaluate the Discriminator Realism Gate to ensure $0.60 \leq \text{AUC}(d_\phi) \leq 0.80$ and clip weights at the 95th percentile to protect Kish’s Effective Sample Size ($\text{ESS} = \frac{(\sum w)^2}{\sum w^2} = 314.5$). For query spot x_{n+1} , the weighted $(1 - \alpha)$ quantile $\hat{q}_\alpha(x_{n+1})$ defines an approximately valid prediction set (Angelopoulos and Bates, 2021):

$$C_\alpha(x_{n+1}) = \{y \in \Delta^{K-1} : \mathcal{S}_{\text{TV}}(x_{n+1}, y) \leq \hat{q}_\alpha(x_{n+1})\}. \quad (1)$$

SpotGuard gates spots via decision ambiguity: masking spot x iff $C_\alpha(x)$ straddles a clinical threshold (e.g., CD8⁺ infiltration > 5%).

System Architecture & Deployment Pipeline. SpotGuard couples a Python engine with a zero-dependency interactive Canvas frontend hosted via GitHub Pages, achieving sub-16ms (> 60 fps) recomputation during slider interaction.

3. Results

Controlled Shift Benchmark Evaluation. We evaluate SpotGuard on a controlled simulation benchmark ($G = 300$ genes, $K = 5$ cell types, $N_{\text{cells}} = 1,000$ per donor, $N_{\text{spots}} = 500$ per tissue) that induces cross-donor and technical platform shift by construction, allowing ground-truth proportions y^* to be known exactly (Table 1).

Baseline Analysis & Shift Performance. The Bayesian Joint CI is an unadjusted product of marginal posterior intervals, which collapses geometrically on Δ^{K-1} for moderate K , while marginal

CI’s achieve 85.4% coverage. Under mild cross-donor shift, weighting matches unweighted coverage (89.8%) at a slight efficiency cost ($\mathcal{V} = 0.058$ vs. 0.054). Under cross-platform shift, SpotGuard prevents a 5.2% coverage drop (89.4% vs. 84.6%), achieving an AURC of 0.138.

Planned Clinical Validation. We are validating SpotGuard on 10x Visium breast carcinoma sections to test whether conformal gatekeeping prunes artifactual CD8⁺ margin calls against orthogonal multiplex immunofluorescence co-staining.

4. Discussion & Limitations

Development Challenges & Limitations. Key challenges addressed include: (1) *Boundary Singularities*: Euclidean and log-ratio losses distort predictions near simplex faces, resolved by Total Variation geometry; (2) *Density Ratios*: PCA projections stabilize discriminator training; (3) *Residual Shift*: Sim-to-real capture variations introduce mild concept shift, rendering coverage approximately valid.

Clinical Rationale & Roadmap. Scalar TV bounds and binary ambiguity masks provide actionable clinical clarity over high-dimensional credible tensors. Future work extends SpotGuard to subcellular platforms (10x Xenium, CosMx).

5. Demo Video & System Access

SpotGuard is publicly available with live interactive demonstrations and full source code:

- **Interactive Web App:** <https://the-mathking.github.io/ML4HDemo/>
- **2-Minute Demo Video:** <https://youtu.be/spotguard-demo>
- **Open-Source Code:** <https://github.com/The-MathKing/ML4HDemo>

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